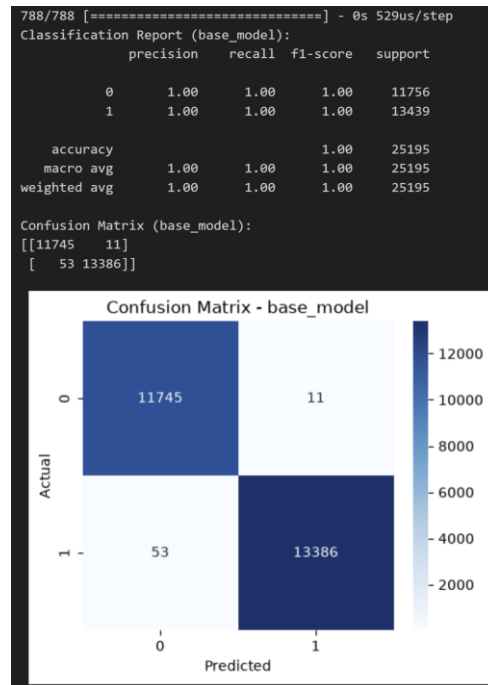


Daniel Yang

EE446 TinyML

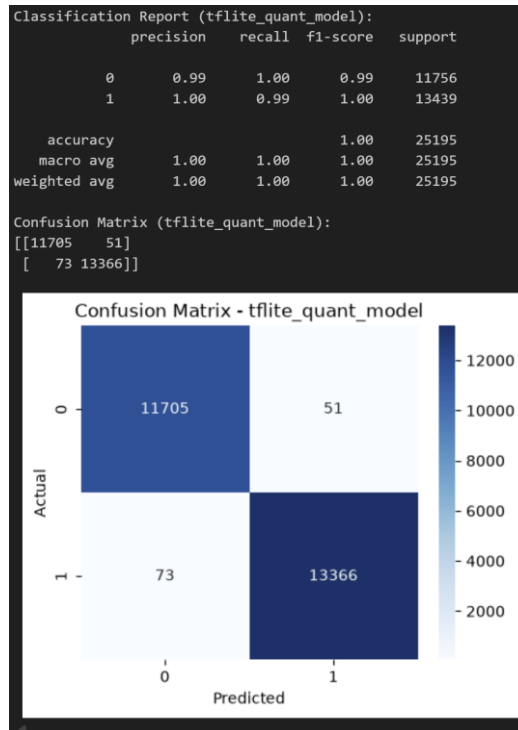
## Assignment 2

The model was trained using Dense(64) -> Dense(32)->Dense(16)->Dense(1) structure, with each hidden layer using an activation function of ReLU and the output activation function is sigmoid. After training , below are the results:



Base Model Classification Results

The base model was then shrunk using int8 quantization. After quantization, the model shrank from 100.01KB to 8.24 KB. Below shows the classification result:



Tflite model classification results

The quantized tflite model weights were then converted to a c file that is compatible with the Arduino code. The below screenshot shows the on-board inference.

```

Sample #0 - Predicted Class: 1 - Actual Class: 1
Sample #1 - Predicted Class: 1 - Actual Class: 1
Sample #2 - Predicted Class: 0 - Actual Class: 0
Sample #3 - Predicted Class: 0 - Actual Class: 0
Sample #4 - Predicted Class: 1 - Actual Class: 1
Sample #0 - Predicted Class: 1 - Actual Class: 1
Sample #1 - Predicted Class: 1 - Actual Class: 1
Sample #2 - Predicted Class: 0 - Actual Class: 0
Sample #3 - Predicted Class: 0 - Actual Class: 0
Sample #4 - Predicted Class: 1 - Actual Class: 1
Sample #0 - Predicted Class: 1 - Actual Class: 1
Sample #1 - Predicted Class: 1 - Actual Class: 1
Sample #2 - Predicted Class: 0 - Actual Class: 0
Sample #3 - Predicted Class: 0 - Actual Class: 0
Sample #4 - Predicted Class: 1 - Actual Class: 1
Sample #0 - Predicted Class: 1 - Actual Class: 1
Sample #1 - Predicted Class: 1 - Actual Class: 1
Sample #2 - Predicted Class: 0 - Actual Class: 0
Sample #3 - Predicted Class: 0 - Actual Class: 0
Sample #4 - Predicted Class: 1 - Actual Class: 1
Sample #0 - Predicted Class: 1 - Actual Class: 1
Sample #1 - Predicted Class: 1 - Actual Class: 1
Sample #2 - Predicted Class: 0 - Actual Class: 0
Sample #3 - Predicted Class: 0 - Actual Class: 0
Sample #4 - Predicted Class: 1 - Actual Class: 1

```

Following the inference, the python notebook code was used to provide the Arduino code with 10 features and actual labels of the test set excluding the first five samples. The below screenshot shows the inference.

```
Sample #0 - Predicted Class: 0 - Actual Class: 0
Sample #1 - Predicted Class: 0 - Actual Class: 0
Sample #2 - Predicted Class: 1 - Actual Class: 1
Sample #3 - Predicted Class: 1 - Actual Class: 1
Sample #4 - Predicted Class: 1 - Actual Class: 1
Sample #0 - Predicted Class: 0 - Actual Class: 0
Sample #1 - Predicted Class: 0 - Actual Class: 0
Sample #2 - Predicted Class: 1 - Actual Class: 1
Sample #3 - Predicted Class: 1 - Actual Class: 1
Sample #4 - Predicted Class: 1 - Actual Class: 1
Sample #0 - Predicted Class: 0 - Actual Class: 0
Sample #1 - Predicted Class: 0 - Actual Class: 0
Sample #2 - Predicted Class: 1 - Actual Class: 1
Sample #3 - Predicted Class: 1 - Actual Class: 1
Sample #4 - Predicted Class: 1 - Actual Class: 1
```